

Mini Project
On
Door Locking System
BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING (IOT)

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Introduction:

An automated door unlocking system simulated in Proteus offers a practical platform for learning about embedded systems and security protocols, allowing users to design and test virtual prototypes before physical implementation. The simulation typically involves a microcontroller (likely a PIC, labeled U1). The system utilizes a 4x4 keypad for user input, allowing for PIN code entry, and an LCD display (LCD1) for visual feedback. The core functionality revolves around the microcontroller processing keypad inputs, comparing them against a stored access code, and subsequently controlling a DC motor (connected via a motor driver, U2) which simulates the door's locking mechanism. The simulation highlights the integration of user interface elements (keypad and LCD), a microcontroller for logic processing, and an actuator (DC motor) for physical action, all within the Proteus environment. This setup provides a practical platform for learning and experimenting with embedded systems and access control principles before physical implementation.

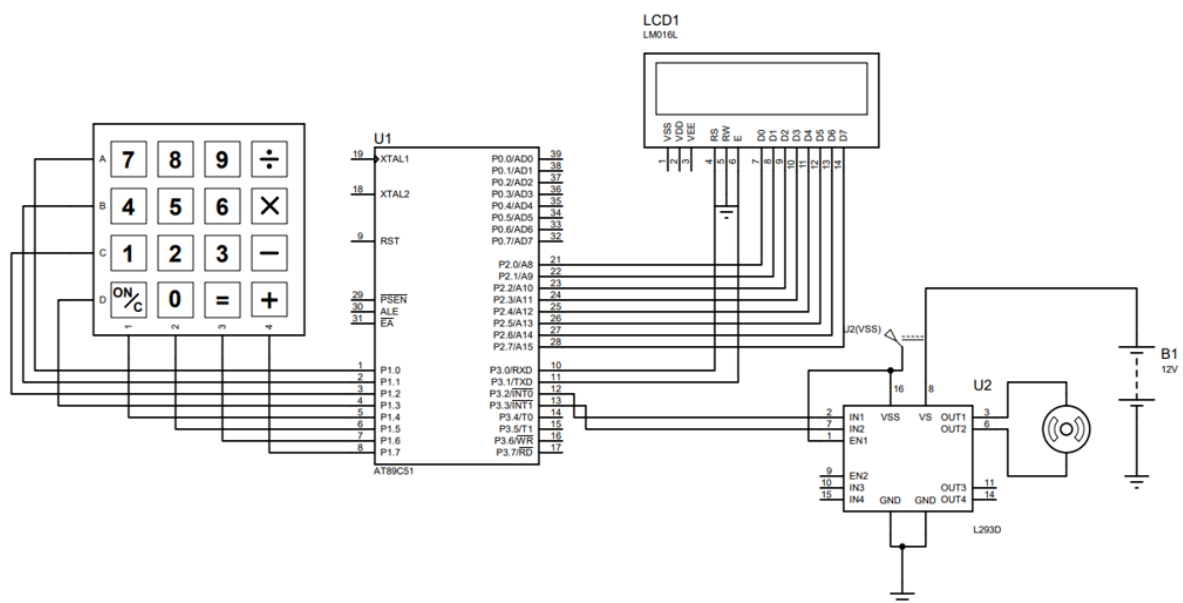
Objective:

This Proteus simulation demonstrates a keypad-based door unlocking system, utilizing a microcontroller to process PIN input and control a DC motor for simulated locking, with LCD feedback. It serves as an educational tool for embedded systems and basic access control principles.

System Components with Diagram:

Component	Description
AT89C51 (Microcontroller)	An 8-bit microcontroller with 4K Flash memory, commonly used for embedded applications.
LCD1 (LM016L) (16x2 LCD)	A 16x2 character LCD display used for visual output in embedded systems.
B1 (12V Battery)	A power source that provides 12V DC for circuit operation.
GND (Ground)	The reference point in an electrical circuit for voltage levels.
(L293D) – Motor Driver IC	A dual H-Bridge motor driver used to control DC motors.
KEYPAD - SMALLCALC	A small-sized calculator-type keypad for user input.

Simulation Diagram:



Implementation Process:

- ❑ **User Input:** The user enters a PIN code using the 4x4 keypad. Each key press generates a unique signal that is sent to the microcontroller (U1).
- ❑ **Microcontroller Processing:**
 - The microcontroller reads the keypad input and stores the entered digits.
 - It compares the entered PIN code with a pre-stored or programmed correct PIN code.
 - If the entered PIN matches the correct PIN, the microcontroller proceeds to the unlocking sequence.
 - If the PIN is incorrect, it may display an error message on the LCD (LCD1) and/or deny access.
- ❑ **Actuator Control:**
 - Upon successful PIN verification, the microcontroller sends a signal to the motor driver (U2).
 - The motor driver amplifies the signal and controls the DC motor.
 - The DC motor rotates, simulating the unlocking of the door's locking mechanism.
- ❑ **User Feedback:**
 - The LCD display (LCD1) provides feedback to the user throughout the process.

- It may display prompts for PIN entry, confirmation of a correct PIN, error messages for incorrect PINs, or status messages indicating whether the door is locked or unlocked.

□ **Power Supply:**

- The B1 component in the circuit provides the necessary power to the circuit.

Code Implementation:

```
#include<reg51.h>
#include<string.h>
sbit RS = P3^0;
sbit EN = P3^1;
sbit IN1 =P3^2;
sbit IN2 = P3^3;
void delay(int a)
{
    int i,j;
    for(i=0;i<a;i++)
    for(j=0;j<255;j++);
}
void cmd(char cm)
{
    P2 = cm;
    RS = 0;
    EN = 1;
    delay(1);
    EN = 0;
}
void dat(char dt)
{
    P2 = dt;
    RS = 1;
    EN = 1;
    delay(1);
    EN = 0;
}

void display(char *lcd)
{
    while(*lcd != '\0')
    {
        dat(*lcd);
        lcd++;
    }
}

void lcdint()
{
    cmd(0x01);
    cmd(0x38);
    cmd(0x0E);
    cmd(0x80);
}

void main()
{
    char pass[5] = "1234";
    char pass2[5];
    int i=0;
    char *ptr;
    ptr = pass2;
    lcdint();
    display("Password-");
    pass2[4]='\0';

    while(1)
    {
        while(i<4)
        {
            P1=0xFE;

```

```

if(P1==0xEE)
{
    *(ptr+i)='7';
    dat('7');
    delay(200);
    cmd(0x06);

    i++;
}
else if(P1==0xDE)
{
    *(ptr+i)='8';
    dat('8');
    delay(200);
    cmd(0x06);

    i++;
}
else if(P1==0xBE)
{
    *(ptr+i)='9';
    dat('9');
    delay(200);
    cmd(0x06);

    i++;
}
else if(P1==0x7E)
{
    *(ptr+i)='/';
    dat('/');
    delay(200);
    cmd(0x06);

    i++;
}

```

```

P1=0xFD;
if(P1==0xED)
{
    *(ptr+i)='4';
    dat('4');
    delay(200);
    cmd(0x06);

    i++;
}
else if(P1==0xDD)
{
    *(ptr+i)='5';
    dat('5');
    delay(200);
    cmd(0x06);

    i++;
}
else if(P1==0xBD)
{
    *(ptr+i)='6';
    dat('6');
    delay(200);
    cmd(0x06);

    i++;
}
else if (P1==0x7D)
{
    *(ptr+i)='*';
    dat('*');
    delay(200);
    cmd(0x06);

    i++;
}

```

```

P1=0xFB;

```

```

if(P1==0xEB)
{
    *(ptr+i)='1';
    dat('1');
    delay(200);
    cmd(0x06);

    i++;
}
else if(P1==0xDB)
{
    *(ptr+i)='2';
    dat('2');
    delay(200);
    cmd(0x06);

    i++;
}
else if(P1==0xBB)
{
    *(ptr+i)='3';
    dat('3');
    delay(200);
    cmd(0x06);

    i++;
}
else if(P1==0x7B)
{
    *(ptr+i)='-';
    dat('-');
    delay(200);
    cmd(0x06);

    i++;
}

```

P1=0xF7;

```

if(P1==0xE7)
{
    *(ptr+i)='C';
    dat('C');
    delay(200);
    cmd(0x06);

    i++;
}
else if(P1==0xD7)
{
    *(ptr+i)='0';
    dat('0');
    delay(200);
    cmd(0x06);

    i++;
}
else if(P1==0xB7)
{
    *(ptr+i)='=';
    dat('=');
    delay(200);
    cmd(0x06);

    i++;
}
else if(P1==0x77)
{
    *(ptr+i)='+';
    dat('+');
    delay(200);
    cmd(0x06);

    i++;
}
}

```

while(i==4)

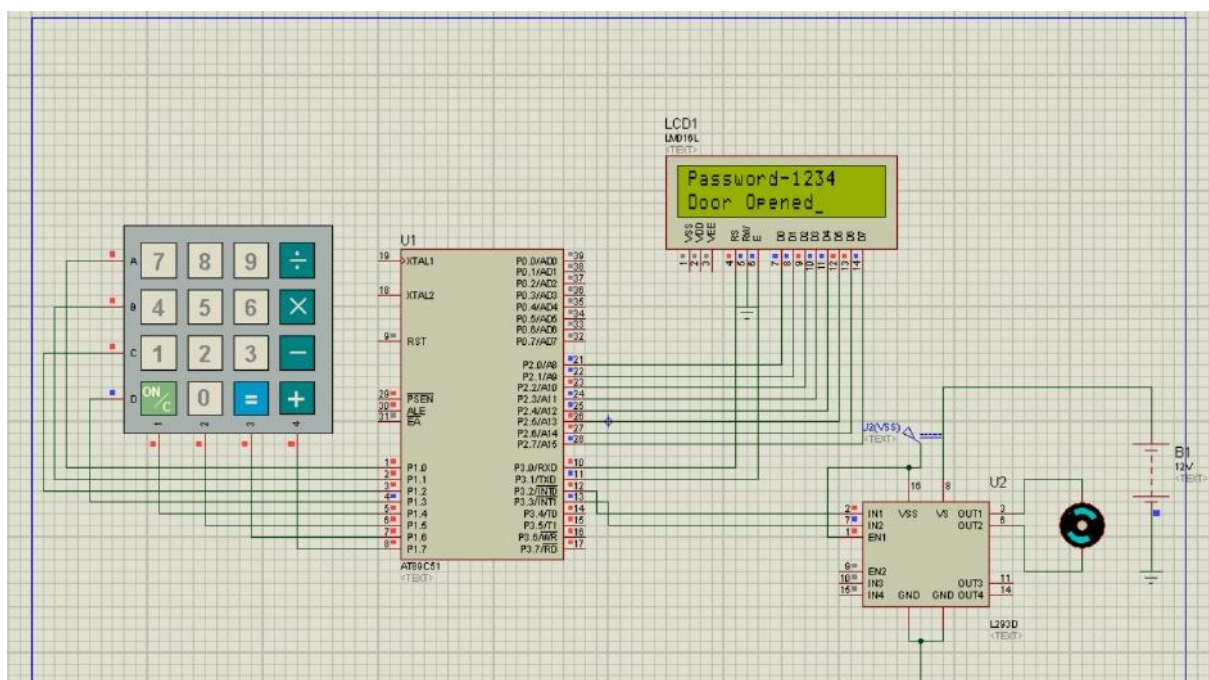
```

{
if ((strcmp(pass, pass2)) == 0)
{
cmd(0xC0);
display("Door Opened");
IN1 = 1;
IN2 = 0;
delay(100);
}
else
{
cmd(0xC0);
display("Incorrect!");
IN1 = 0;
IN2 = 0;
delay(100);
}
}

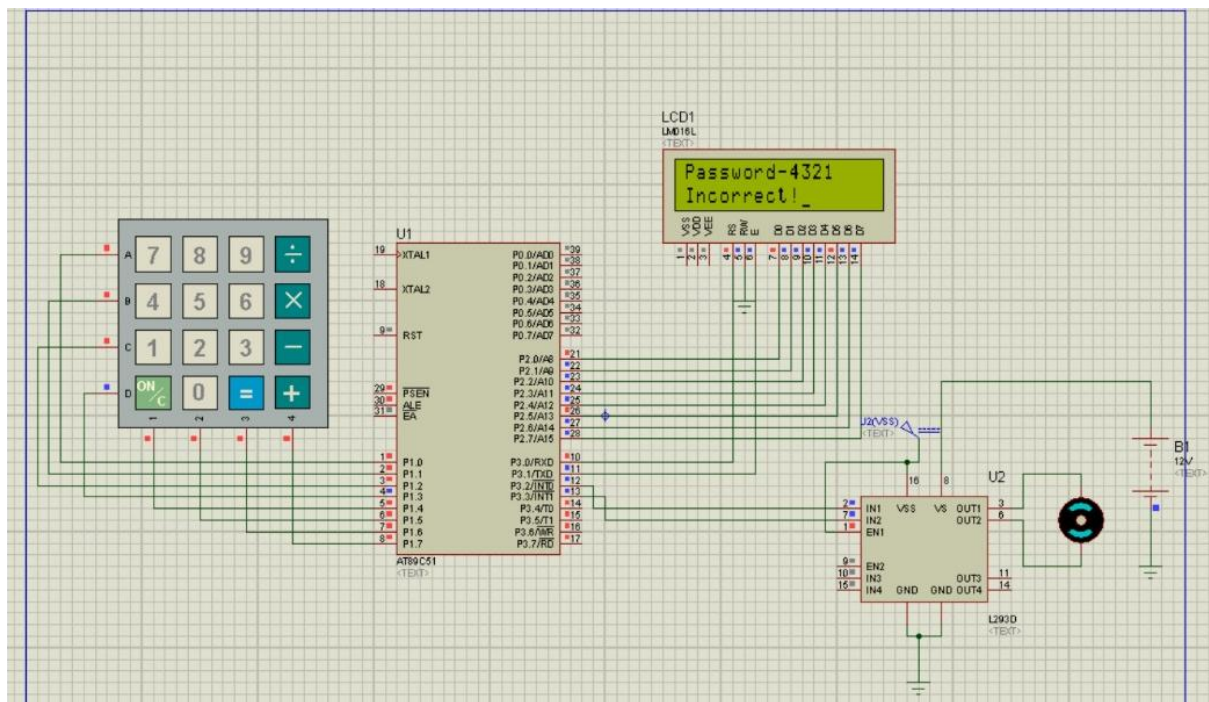
```

Simulation Results:

- Success case



- Failure case



Conclusion:

The Proteus simulation effectively realizes a keypad-controlled door unlocking system, showcasing the integration of a microcontroller, 4x4 keypad input, LCD display for feedback, and a DC motor to simulate the locking mechanism. Critically, the system is designed to unlock the simulated door only when the correct PIN code is entered via the keypad; otherwise, access is denied. This project provides a practical demonstration of embedded system principles, particularly microcontroller programming and interfacing, within the context of a basic security access system. It serves as a valuable educational exercise for understanding how user input can be processed to control a physical output based on conditional logic.